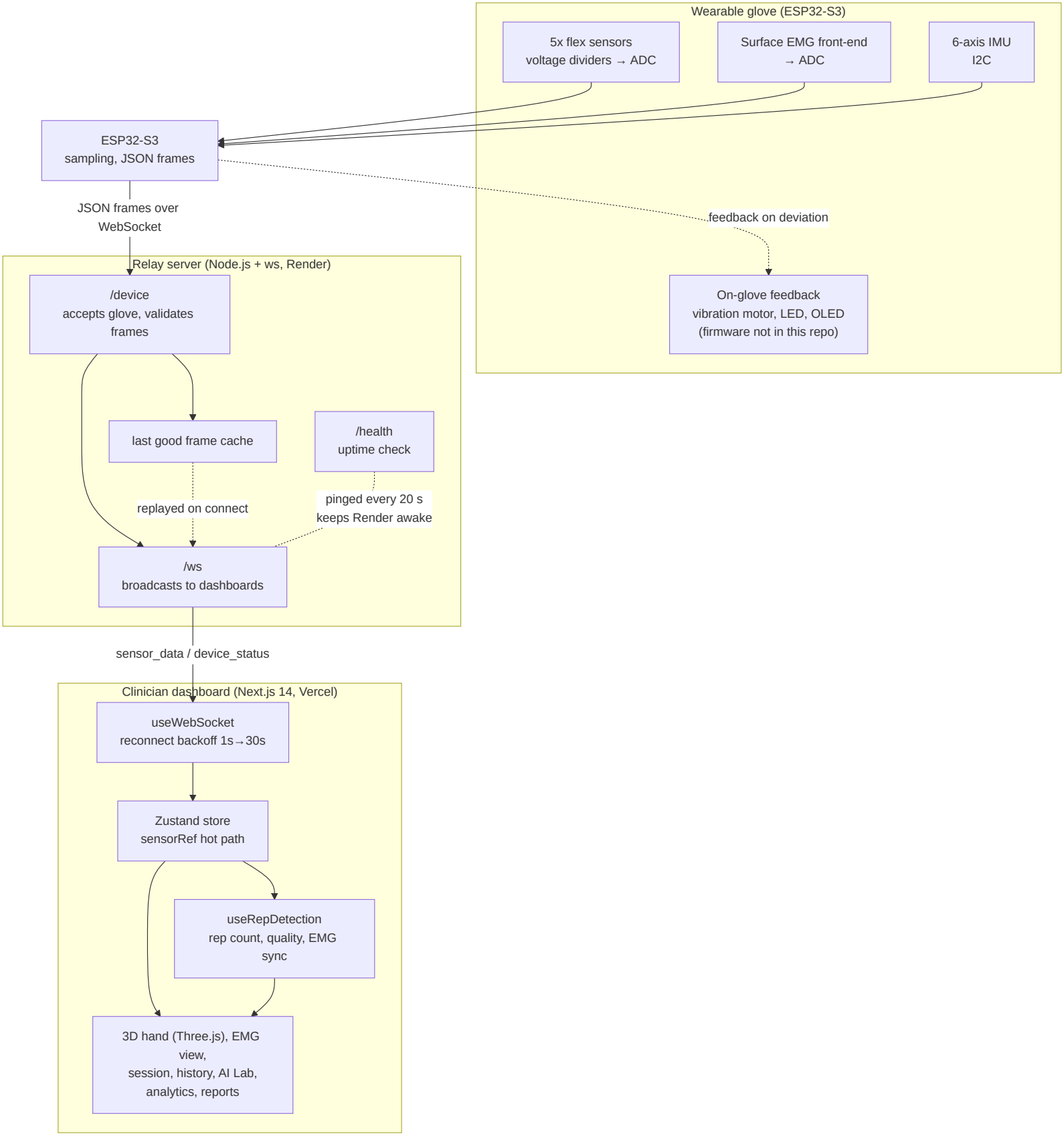
Wearable glove (ESP32-S3)
5x flex sensors
voltage dividers → ADC
Surface EMG front-end
→ ADC
6-axis IMU
I2C
ESP32-S3
sampling, JSON frames
JSON frames over
WebSocket
feedback on deviation
On-glove feedback
vibration motor, LED, OLED
(firmware not in this repo)
Relay server (Node.js + ws, Render)
/device
accepts glove, validates
frames
last good frame cache
/health
uptime check
replayed on connect
pinged every 20 s
keeps Render awake
/ws
broadcasts to dashboards
sensor_data / device_status
Clinician dashboard (Next.js 14, Vercel)
useWebSocket
reconnect backoff 1s→30s
Zustand store
sensorRef hot path
useRepDetection
rep count, quality, EMG
sync
3D hand (Three.js), EMG
view,
session, history, AI Lab,
analytics, reports